



Engineering PEG-fatty acid stapled, long-acting peptide agonists for G protein-coupled receptors

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Abstract

G protein-coupled receptors (GPCRs) play a key role in signal transduction and human pathophysiological processes. Family B GPCRs are activated by a number of secreted peptide hormones, and engineering of these peptide ligands in order to improve stability and half-life, and therefore clinical efficacy has proven successful for drug discovery. In this chapter we discuss a novel peptide engineering strategy that combines peptide side chain stapling with covalent incorporation of a serum protein binding motif in a single step. The application of this approach to the enhancement of the helicity and stability of GLP-1R peptide agonists, resulting in their improved in vitro potencies, in vivo half-lives and ultimately efficacies, will be described. Discussion of the stapling technology and target selection rationale, peptide engineering and final biological characterization of the long-acting agonists will also be provided.

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1. Introduction

1.1 GPCRs and their important physiopathological roles

G protein-coupled receptors (GPCRs) are membrane-bound proteins that have seven transmembrane domains linked by three intracellular and three extracellular loops. Their ligand-binding sites are highly specialized so that each receptor responds only to a limited variety of chemicals which bind with high affinity. Examples of GPCR ligands are peptides, proteins, lipid-derived molecules, small organic compounds and ions (Marinissen & Gutkind, 2001). GPCRs have been of long-standing interest as pharmaceutical drug targets, as they are involved in a plethora of pathophysiological processes, including the regulation of neuronal excitability, metabolism, reproduction, hormonal homeostasis, and behavior (Vassilatis et al., 2003). It is estimated that around 34% of all Food and Drug Administration (FDA) approved drugs target 108 members of the GPCR family (Hauser et al., 2018).

GPCRs are generally classified into multiple superfamilies. Family B GPCRs, or the so-called secretin receptor family, are a small but structurally and functionally diverse set of receptors. These proteins are vital to many physiological functions and serve as key drug targets for several human diseases such as type II diabetes mellitus (T2DM), migraine, osteoporosis, depression, and anxiety (Amso, Cornish, & Brimble, 2016; Catapano & Manji, 2007; Persaud, 2017; Szafran et al., 2013). Members of this family include receptors for polypeptide hormones of 27–141 residues in length. Nine of these receptors are targeted by ligands that are structurally related to one another, examples of which include glucagon-like peptides (GLP-1 and GLP-2), glucagon, glucose-dependent insulintropic polypeptide (GIP), vasoactive intestinal peptide (VIP), pituitary adenylate cyclase-activating polypeptide (PACAP) and growth hormone-releasing hormone (GHRH). These are summarized in Table 1.

1.2 Engineering peptide GLP-1R ligands for the treatment of T2DM

Currently, there are more than 30 GPCRs that have been associated with the progression of T2DM. However, only GLP-1R has been successfully targeted for anti-diabetic therapy (Ahren, 2009; Berger et al., 2015). GLP-1R can be targeted by injectable GLP-1 mimetics as well as by circulating active GLP-1, but has proved to be a difficult target for developing small molecule ligands that can be administered in oral formulations (Willard, Bueno, & Sloop, 2012).

Table 1 Examples of hormone receptors in family B GPCRs.

Receptor	Ligand	Function
Glucagon-like peptide 1 receptor (GLP-1R)	GLP-1	↑ Insulin and satiety ↓ Glucagon and gastric emptying
Glucagon-like peptide 2 receptor (GLP-2R)	GLP-2	↑ Proliferation of intestinal mucosal villus epithelium
Gastric inhibitory polypeptide receptor (GIPR)	GIP	↑ Insulin, glucagon and fat storage
Glucagon receptor (GCGR)	Glucagon	↑ Insulin and energy expenditure ↓ Food intake
Parathyroid hormone 1 receptor (PTH1R)	Parathyroid hormone, PTH-related peptide (PTHrP)	↑ Osteoblast proliferation and formation of new bone – Maintenance of calcium and phosphate homeostasis

Glucagon-like peptide 1 (GLP-1) is a naturally-occurring incretin hormone released into the circulation by the L cells of the gut in response to ingested nutrients. GLP-1 has various physiological effects that make it an attractive candidate for T2DM therapy. By binding to its cognate receptor (GLP-1R) GLP-1 is able to promote insulin secretion while suppressing glucagon secretion, but only when glucose levels are raised, thus offering the potential to lower plasma glucose levels while reducing the risk of hypoglycemia (Baggio & Drucker, 2007; Nauck et al., 1993). Furthermore, GLP-1 decreases the rate of gastric emptying, and reduces appetite, thus resulting in weight loss (Kreymann, Williams, Ghatei, & Bloom, 1987; Meier et al., 2003). In both in vitro and rodent studies, GLP-1 displays proliferative and anti-apoptotic properties in pancreatic β cells that could potentially be exploited to enhance β cell mass, although the translatability of these findings to humans remains uncertain (Brubaker & Drucker, 2004).

Despite the great potential of GLP-1 as a therapeutic for T2DM, the peptide is rapidly degraded and inactivated by the enzyme dipeptidyl peptidase-4 (DPP-4), which results in a short in vivo half-life of about 1.5 min (Drucker & Nauck, 2006). Consequently, the native peptide is only effective if administered via continuous infusion and is therefore impractical as a diabetes medication (Deacon, 2007; Larsen, Hylleberg, Ng, & Damsbo, 2001). Hence, efforts toward therapeutic targeting of GLP-1R have focused on GLP-1 peptide *mimetics* resistant to enzymatic degradation, and with prolonged plasma half-life.

Several peptide drugs targeting GLP-1R have been approved for the treatment of T2DM in the United States: exenatide, liraglutide, lixisenatide, albiglutide, dulaglutide and semaglutide (Oh & Olefsky, 2016). A brief discussion of each is given in Sections 1.3 and 1.4. They are all mimetics of the naturally-occurring GLP-1 peptide and cause improved insulin release and reduced glucagon release in a glucose-dependent manner following receptor binding, with a consequently low risk of hypoglycemia. Their effects on GLP-1 receptors in the central nervous system and gastrointestinal tract also lead to reduced appetite and delayed glucose absorption, with concomitant weight loss (Donnelly, 2012).

The key structural differences between the GLP-1 analogs summarized are illustrated in Fig. 1. These variances significantly affect product half-life and dosing interval, and as a result the peptides are classified as either short- or long-acting agonists. Furthermore, there is an important correlation in

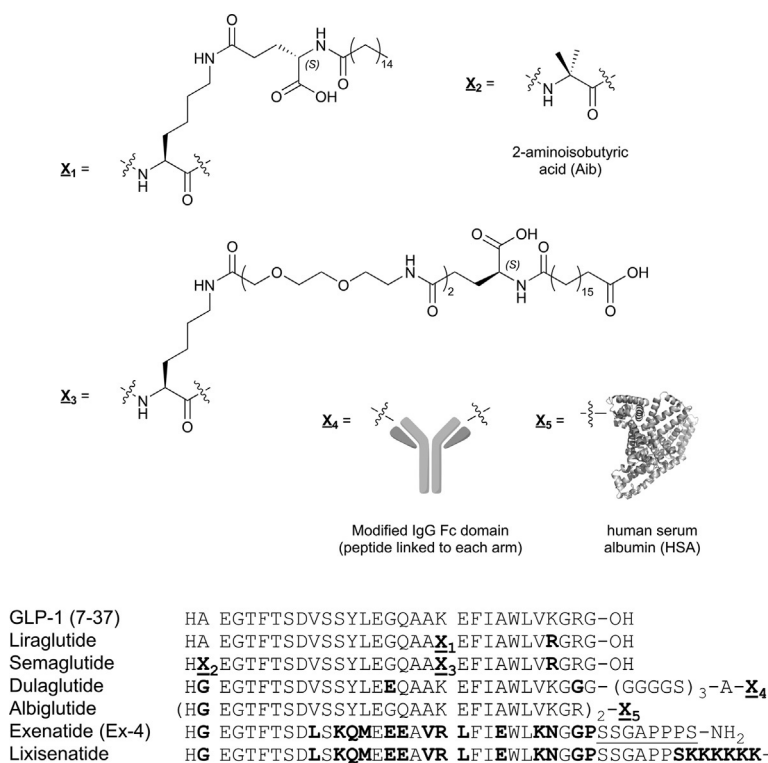


Fig. 1 Structures of GLP-1 analog approved drugs (Cheang & Moyle, 2018; Glaesner et al., 2010).

the GLP-1R agonist class of therapeutics between pharmacokinetic properties and efficacy. While short-acting agonists display a marked ability to reduce gastric emptying, their long-acting counterparts with pharmacologically relevant exposure more than 24 h have less influence on gastric emptying but better glucose lowering effect (Meier, 2012).

1.3 Short-acting GLP-1 agonists

1.3.1 Exenatide (Byetta[®], AstraZeneca)

Exenatide was originally isolated from the saliva of the lizard *Heloderma suspectum* (Gila monster) and became the first GLP-1R agonist to be approved by the FDA for the treatment of T2DM in 2005. This non-human peptide shares 53% homology with human GLP-1 and was shown to have a longer plasma half-life (2–3 h) than GLP-1 and >1000-fold increased potency in lowering glucose levels than GLP-1 (Lam & See, 2006). The reason for the longer duration of action is the presence of His-Gly at the N-terminus (instead of His-Ala in the case of GLP-1), which results in resistance to degradation by DPP-4. Consequently, exenatide is approved to be administered twice-daily by subcutaneous injection (de Graaf et al., 2016).

In phase III clinical trials, patients treated with exenatide exhibited a significant reduction in glycated hemoglobin (Hb1Ac) compared with the placebo group (Vilsboll, Agerso, Krarup, & Holst, 2003). Also, head-to-head studies have shown that twice-daily administration of exenatide resulted in a significant increase in reduction in body weight when compared to lixisenatide and dulaglutide (Madsbad, 2016; Trujillo, Nuffer, & Ellis, 2015).

1.3.2 Lixisenatide (Lyxumia[™]/Adlyxin[™], Sanofi/Zealand Pharma)

Lixisenatide, an analog of exenatide, was approved as a once-daily injection in 2009. The native exenatide sequence is extended to incorporate six extra lysine residues at the C-terminus, with deletion of one C-terminal proline. This modification slightly increases the half-life of exenatide to 3–4 h, allowing for once-daily subcutaneous administration, and importantly results in a fourfold increase in GLP-1 receptor binding (McCarty, Coleman, & Boland, 2017).

Clinical efficacy of lixisenatide compared to twice-daily exenatide has been assessed in a randomized open label, actively controlled, 24-week trial in T2DM patients. In this study, 20 µg lixisenatide demonstrated noninferior improvements in HbA1c, with slightly lowered mean weight loss, but better gastrointestinal tolerability and lower incidence of hypoglycemia compared with twice-daily 10 µg exenatide (Rosenstock et al., 2013).

1.3.3 Liraglutide (Victoza[®], Novo Nordisk)

Liraglutide is a GLP-1R agonist which was approved by the FDA as a once-daily injection for the treatment of T2DM in 2010 and for obesity in 2014. It has 97% homology to human GLP-1(7-37), with the inclusion of two modifications. The first is a Lys-to-Arg amino acid substitution at position 34, and the other is the conjugation of a 16-carbon acyl moiety to the side chain of Lys-26 via a γ -glutamic acid spacer (Agersø, Jensen, Elbrønd, Rolan, & Zdravkovic, 2002). The attached fatty acid moiety fosters binding to serum albumin, thereby delaying renal excretion (Madsen et al., 2007). Consequently, liraglutide has a half-life of approximately 13 h in humans (Elbrønd et al., 2002; Knudsen et al., 2000).

In clinical trials, liraglutide was able to dose-dependently decrease HbA1c, and both fasting and postprandial plasma glucose levels. Furthermore, the peptide was shown to enhance insulin secretion, improve surrogate measures of β cell function, and improve some biomarkers of cardiovascular risk in patients with T2DM (Perry, 2011; Scott, 2014). Moreover, daily 3 mg liraglutide exhibited significant reductions in body weight in obese patients without T2DM when compared to placebo (Pi-Sunyer et al., 2015).

1.4 Long-acting GLP-1 agonists

1.4.1 Albiglutide (Tanzeum[™], GlaxoSmithKline)

Albiglutide is a long-acting GLP-1 analog which was approved by the FDA in 2014. It is given as a once-weekly subcutaneous injection. Albiglutide consists of two GLP-1(7-36) molecules fused in tandem to human serum albumin. The peptide sequence has a Gly-to-Ala replacement at position 8, which confers enhanced resistance to DPP-4-mediated protein aggregation. Additionally, fusion of the peptide to albumin results in half-life extension of 6–8 days in humans (Bush et al., 2009).

Despite being a bulky fusion protein, albiglutide still possesses very high binding affinity to GLP-1R (Manandhar & Ahn, 2015). In the HARMONY clinical trials albiglutide lowered HbA1c levels when compared with pioglitazone and insulin lispro, suggesting suitability as a valid alternative to insulin lispro as an add-on to basal insulin (Lindamood & Taylor, 2015). However, in the HARMONY 7 trial, albiglutide was shown to be less efficacious than liraglutide with respect to weight loss (Pratley et al., 2014).

1.4.2 Exenatide LAR (Bydureon[™], Amylin Pharmaceuticals/Bristol-Myers Squibb)

Exenatide long-acting release (LAR) is a novel formulation of exenatide which was approved as a once-weekly injection for the treatment of

T2DM in 2012. In the formulation, exenatide is non-covalently entrapped within a poly(D,L-lactic-co-glycolic acid) matrix to form microspheres. This results in slow release of the peptide and controlled delivery (DeYoung, MacConell, Sarin, Trautmann, & Herbert, 2011). The microspheres allow steady-state concentrations to be achieved in 6–10 weeks, providing a median plasma half-life for exenatide of approximately 2 weeks (Drucker et al., 2008).

One clinical trial showed a significantly greater reduction of HbA1c after 30 weeks in individuals treated with 2 mg of once-weekly exenatide LAR compared to those who received 10 µg of twice-daily exenatide (Drucker et al., 2008). In addition, a significant reduction in body weight was observed with patients treated with 2 mg exenatide LAR compared to placebo (Kim et al., 2007).

1.4.3 Dulaglutide (Trulicity™, Eli Lilly)

Dulaglutide is a once-weekly GLP-1R agonist approved for the treatment of T2DM in 2014. It is a recombinant DNA-produced polypeptide analog of GLP-1(7–37) that is covalently linked to each Fc arm of human immunoglobulin G4 (IgG4). The structure of dulaglutide confers improved solubility, reduced immunogenicity, and reduced rate of renal clearance (Wysham et al., 2015). The peptide can be used either as a stand-alone therapy or in combination with other medicines for T2DM, in particular metformin and insulin.

Dulaglutide has been studied in comparison with other anti-diabetic drugs in multiple trials. In the AWARD-6 trial, once-weekly 1.5 mg dulaglutide showed a comparable efficacy to daily 1.8 mg liraglutide in reducing HbA1c levels, but was less efficacious than liraglutide with respect to weight loss (Dungan et al., 2014).

1.4.4 Semaglutide (Ozempic®, Novo Nordisk)

Semaglutide is the newest GLP-1R agonist, and was approved by the FDA as a once-weekly injection for the treatment of T2DM in late 2017. The peptide is chemically similar to liraglutide, with the inclusion of two structural modifications (Lau, Bloch, et al., 2015). The first is replacement of Gly with the non-proteinogenic amino acid 2-aminoisobutyric acid (Aib) at position 2. The second is the attachment of octadecanoic diacid to the side chain of Lys-26 through a short polyethylene glycol (PEG) spacer and a γ-glutamic acid linker. The amino acid substitution renders the peptide more resistant to DPP-4 compared to liraglutide, and the presence of the 18-carbon fatty acid moiety results in a high binding affinity for serum

albumin, which translates to a half-life of approximately 7 days in humans (van Witteloostuijn, Pedersen, & Jensen, 2016).

In a 12-week phase II study, semaglutide lowered HbA1c levels and body weight, with higher doses (0.8 and 1.6 mg) being more effective than liraglutide (1.2 or 1.8 mg). A once-daily oral formulation of semaglutide has also been developed which uses the absorption enhancer sodium *N*-[8-(2-hydroxybenzoyl)amino]caprylate (SNAC), and is currently undergoing phase III clinical trials.



2. Design and synthesis of a PEG-fatty acid stapled, long-acting GLP-1R agonist

2.1 Combined peptide stapling and lipidation

As can be seen in the previous section, lipidation of biologically-active peptides provides a highly effective means of prolonging half-life, and thus transforms potent hits with poor pharmacokinetic properties into lead molecules suitable for drug development. Peptide *stapling* is another strategy that offers promise in peptide drug discovery for the enhancement of peptide potency and proteolytic stability through stabilization of alpha-helical structure (Henchey, Jochim, & Arora, 2008). This approach (illustrated in Fig. 2) involves the attachment of a tether between side chains on the same face of a peptide alpha helix. Peptide stapling has been applied to numerous targets in multiple disease areas, and is a rapidly expanding area of research in peptide drug discovery (Verdine & Hilinski, 2012; White & Craik, 2016).

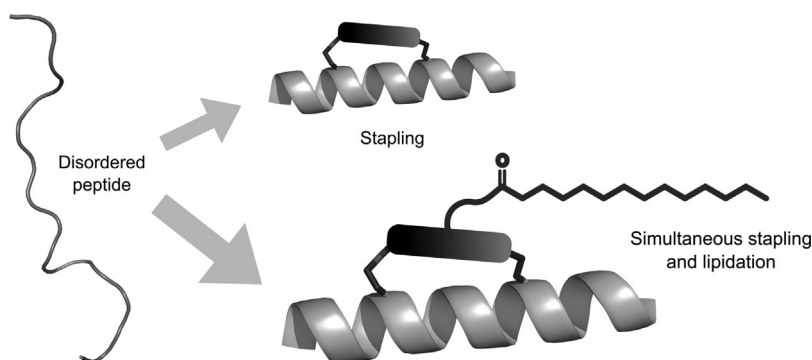


Fig. 2 Principle of simultaneous peptide stapling and lipidation. A covalent tether across side chains stabilizes alpha-helical peptide conformation, enhancing potency and proteolytic stability. Lipidation can be incorporated concurrently via the staple to achieve simultaneous half-life prolongation.

A diverse set of stapling chemistries have been utilized to date (Lau, de Andrade, Wu, & Spring, 2015), including disulfide, thioether and lactam linkages (Jamieson, Boutard, Sabatino, & Lubell, 2013), in addition to “all-hydrocarbon” tethers installed using Grubbs ring-closing metathesis (Cromm, Spiegel, & Grossmann, 2015), an example of which is currently in clinical development (Meric-Bernstam et al., 2017).

The following sections describe the application of a unique simultaneous stapling/fatty acid conjugation strategy, which achieves increased potency, stability and prolonged half-life concurrently in a single chemical step. This account focuses on the generation of stapled, long-acting versions of the GLP-1R agonist exenatide via this approach, and covers sequence engineering, fatty acid optimization and biological characterization of the long-acting analogs.

2.2 Sequence engineering and staple optimization

The Gila monster venom-derived peptide exenatide (Ex-4) serves as an ideal starting point for engineering due to its significant potency at the GLP-1 receptor while simultaneously exhibiting poor intrinsic pharmacokinetic properties. Upon subcutaneous (s.c.) administration Ex-4 has a half-life of only 2–3 h in humans (Meier, 2012). The first step in the design process is the identification of suitable sites on the peptide for covalent modification, which should be selected with reference to structural data if available. The crystal structure of Ex-4 in complex with the GLP-1R extracellular domain is shown in Fig. 3.

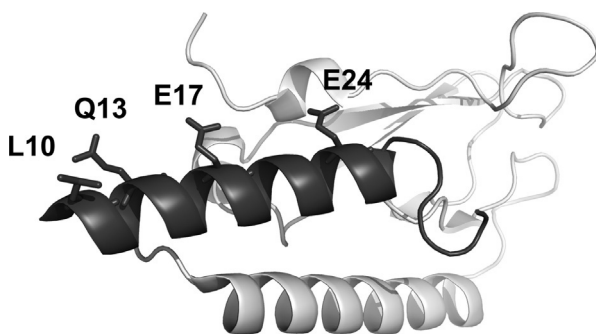


Fig. 3 Crystal structure of Ex-4 (black) in complex with GLP-1R isolated extracellular domain (gray); PDB ID: 3C5T (Runge, Thøgersen, Madsen, Lau, & Rudolph, 2008; The PyMOL Molecular Graphics System, Version 1.6 Schrödinger, LLC). Solvent-exposed residues L10, Q13, E17 and E24 (side chains shown) are positioned on the same face of the helix, away from the receptor interface, and are therefore ideal sites for stapling.



Redrawn from Yang, P.-Y., Zou, H., Chao, E., Sherwood, L., Nunez, V., Keeney, M., et al. (2016). Engineering a long-acting, potent GLP-1 analog for microstructure-based transdermal delivery. *Proceedings of the National Academy of Sciences of the United States of America*, 113, 4140–4145. doi:10.1073/pnas.1601653113.

The Ex-4 sequence is shown in Fig. 4 alongside sequences **A–C**, which incorporate Cys mutations at the stapling positions highlighted in Fig. 3. Representative staple structures are also shown. Staples **4–6** not incorporating any fatty acyl moiety are used first to investigate the effect of stapling alone, without lipid conjugation. Stapling of the Cys-containing peptides is carried out via the solution-phase protocol given in the following table. The linker chain length is varied to identify the optimum macrocycle that matches the side chain spacing on the alpha helix. Linkers **4–6** are particularly useful for scanning of Cys positions across a sequence in an initial screen, especially where structural information is lacking. Biological characterization of the resulting stapled peptides is essential and will be discussed in Section 2.3.

Solution-phase peptide stapling protocol

1. Dissolve peptide with 1.5 equiv. of staple to a final peptide concentration of 2 mM in NH_4HCO_3 buffer (30 mM, pH 8.5) containing 25% acetonitrile. The peptide concentration can be reduced or a higher percentage of acetonitrile added in the case of particularly insoluble peptides.
 2. Readjust to pH 8.5 using ammonia.
 3. Agitate reaction at room temperature for 2–4 h.
 4. Acidify reaction to pH 5 using glacial acetic acid.
 5. Purify directly or lyophilize (depending on reaction volume).
-

Once staple length and position have been optimized with reference to in vitro receptor activity (see below), fatty acyl groups can be incorporated into the staple (as exemplified by staples **1–3**) in order to enhance in vivo half-life. The staples shown can be synthesized using standard peptide coupling chemistry, either via solid- or solution-phase techniques, details of which are given in [Yang et al. \(2016\)](#). It is essential to optimize PEG and fatty acyl chain length and fatty acid type (presence/placement of carboxylate groups, etc.) in order to achieve favorable serum protein binding—and thus optimal pharmacokinetic profile—for the conjugates, without affecting receptor binding potency.

2.3 Biological characterization

When engineering long-acting peptide drug molecules, in vitro potency and in vivo pharmacokinetic (PK) data are key in identifying the most successful candidates, and as such should guide the design and optimization at all stages of discovery. Prior to in vivo evaluation, it is necessary to confirm acceptable potency is retained for stapled analogs in a suitable cell-based receptor activation assay. A cyclic adenosine monophosphate (cAMP) responsive element-driven luciferase reporter system (CRE-luc) is a convenient way to monitor intracellular cAMP synthesis resulting from specific GPCR activation in HEK293 cells overexpressing the GLP-1 receptor ([Yang et al., 2016](#)). Using such an assay system, receptor activation by the stapled peptide conjugates can be evaluated via a simple luminescence readout.

Representative in vitro GLP-1 receptor activity data are given in [Fig. 5](#). Variation of the staple length using non-fatty acyl staples produces analogs **A4** (*i*, *i* + 7 with 10-atom linker) and **B5** (*i*, *i* + 11 with 13-atom linker), which have activities comparable to that of the native Ex-4 sequence. Modification of staple **4** to incorporate the lipid moiety, followed by optimization of

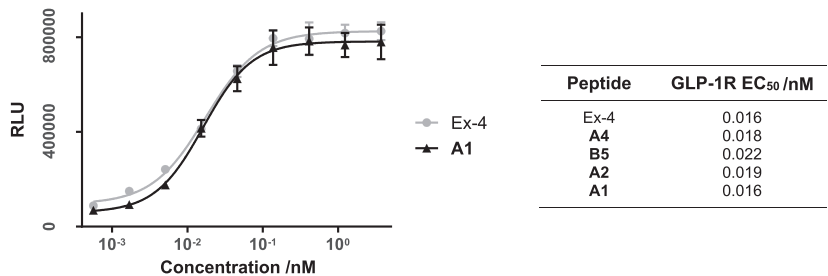


Fig. 5 Dose-response curves and EC₅₀ data for selected stapled agonists (CRE-luc reporter assay for GLP-1R activation). Data from Yang, P. -Y., Zou, H., Chao, E., Sherwood, L., Nunez, V., Keeney, M., et al. (2016). *Engineering a long-acting, potent GLP-1 analog for microstructure-based transdermal delivery*. Proceedings of the National Academy of Sciences of the United States of America, 113, 4140 – 4145. doi:10.1073/pnas.1601653113.

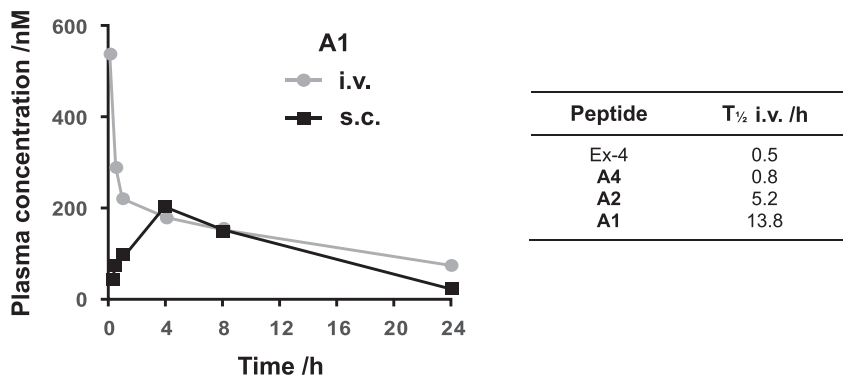


Fig. 6 Mouse plasma concentration of **A1** after i.v. and s.c. injection (40 µg/kg), and i.v. half-lives for selected conjugates. Data from Yang, P. -Y., Zou, H., Chao, E., Sherwood, L., Nunez, V., Keeney, M., et al. (2016). *Engineering a long-acting, potent GLP-1 analog for microstructure-based transdermal delivery*. Proceedings of the National Academy of Sciences of the United States of America, 113, 4140 – 4145. doi:10.1073/pnas.1601653113.

PEG and fatty acid chain length, results in staples **1** and **2**, which retain potent in vitro activity toward the GLP-1 receptor (agonists **A1** and **A2**).

With a set of candidates exhibiting favorable in vitro potency in hand, the next step is to assess their PK profiles in vivo. Mouse PK data for key analogs are shown in Fig. 6. While stapling of Ex-4 alone does not significantly enhance plasma half-life (**A4**), stapling and conjugation to an 18-carbon fatty acid via a PEG₃ linker dramatically reduces in vivo clearance (**A2**), with a demonstrated 10-fold increase in T_{1/2} over the native sequence upon intravenous administration. Furthermore, incorporation of an internal carboxylic acid moiety in addition to the distal carboxylate further increased the half-life to almost 14 h (**A1**), representing a near 30-fold increase. This

impressive rodent half-life is equivalent to that of semaglutide, which is approved as a once-weekly injectable for diabetes care. This result illustrates the power of the approach described in taking peptides with previously poor PK properties and generating long-acting therapeutic candidates suitable for potential once-weekly administration in human.

Translation of the prolonged in vivo half-life of **A1** into long-lasting, superior in vivo efficacy is demonstrated in the chronic 5-week weight loss study in diet-induced obesity (DIO) model mice summarized in Fig. 7. Daily subcutaneous treatment of 25-week-old DIO mice for 5 weeks with **A1** results in a steady reduction in body weight and fasting blood glucose

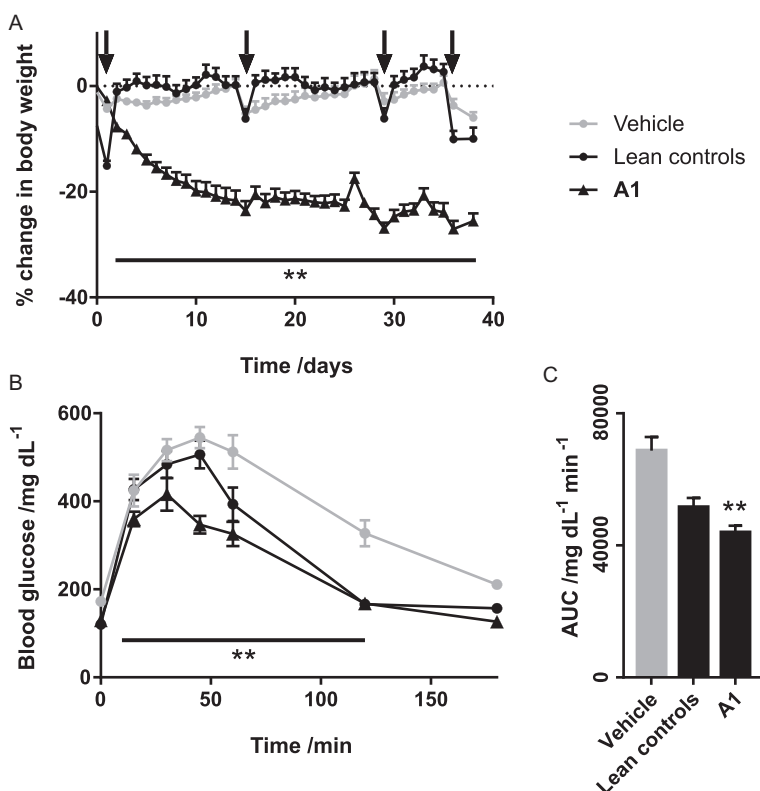


Fig. 7 Demonstration of favorable chronic in vivo efficacy profile for long-acting Ex-4 analog **A1** (40 $\mu\text{g/kg}$): (A) % change in body weight of diet-induced obesity (DIO) model mice during 5-week treatment (arrows indicate fasting for OGTT, see below); (B) oral glucose tolerance test (OGTT) at day 14; (C) area under the curve (AUC) for day 14 OGTT; $**P < 0.01$, **A1** vs vehicle. Adapted from Yang, P. -Y., Zou, H., Chao, E., Sherwood, L., Nunez, V., Keeney, M., et al. (2016). Engineering a long-acting, potent GLP-1 analog for microstructure-based transdermal delivery. *Proceedings of the National Academy of Sciences of the United States of America*, 113, 4140–4145. doi:10.1073/pnas.1601653113.

levels. Significant decreases in cumulative food intake and visceral fat mass also occur concurrently (not shown). Improved glucose control is also observed in an oral glucose tolerance test (OGTT) on day 14. **A1** treatment reduces plasma cholesterol, and plasma and liver triglyceride levels relative to vehicle (not shown). In contrast, Ex-4 treatment alone at 42 µg/kg (daily i.v. dosing for 1 week) does not produce a body weight reduction effect in the DIO mouse model (Yang et al., 2016).



3. Summary

As demonstrated above, the simultaneous application of peptide stapling and lipidation is a unique and powerful approach for generating potent, long-acting analogs of existing biologically active peptides with poor pharmacokinetic performance. This versatile and modular approach can be applied to a wide variety of therapeutic peptides across multiple disease areas, such as GLP-1 (Yang et al., 2016), GLP-2 (Yang et al., 2018), oxyntomodulin (Muppidi et al., 2016), PrRP (unpublished) and PYY (unpublished). Harnessing the combined benefits of peptide stapling and lipidation simultaneously offers the unique advantages of both activity and stability enhancement, and half-life prolongation, producing results not achievable using either technique alone.

References

- Agersø, H., Jensen, L. B., Elbrønd, B., Rolan, P., & Zdravkovic, M. (2002). The pharmacokinetics, pharmacodynamics, safety and tolerability of NN2211, a new long-acting GLP-1 derivative, in healthy men. *Diabetologia*, 45, 195–202. <https://doi.org/10.1007/s00125-001-0719-z>.
- Ahren, B. (2009). Islet G protein-coupled receptors as potential targets for treatment of type 2 diabetes. *Nature Reviews Drug Discovery*, 8, 369–385. <https://doi.org/10.1038/nrd2782>.
- Amso, Z., Cornish, J., & Brimble, M. A. (2016). Short anabolic peptides for bone growth. *Medicinal Research Reviews*, 36, 579–640. <https://doi.org/10.1002/med.21388>.
- Baggio, L. L., & Drucker, D. J. (2007). Biology of incretins: GLP-1 and GIP. *Gastroenterology*, 132, 2131–2157. <https://doi.org/10.1053/j.gastro.2007.03.054>.
- Berger, M., Scheel, D. W., Macias, H., Miyatsuka, T., Kim, H., Hoang, P., et al. (2015). Gα_{i/o}-coupled receptor signaling restricts pancreatic beta-cell expansion. *Proceedings of the National Academy of Sciences of the United States of America*, 112, 2888–2893. <https://doi.org/10.1073/pnas.1319378112>.
- Brubaker, P. L., & Drucker, D. J. (2004). Minireview: Glucagon-like peptides regulate cell proliferation and apoptosis in the pancreas, gut, and central nervous system. *Endocrinology*, 145, 2653–2659. <https://doi.org/10.1210/en.2004-0015>.
- Bush, M. A., Matthews, J. E., De Boever, E. H., Dobbins, R. L., Hodge, R. J., Walker, S. E., et al. (2009). Safety, tolerability, pharmacodynamics and pharmacokinetics of albiglutide, a long-acting glucagon-like peptide-1 mimetic, in healthy subjects. *Diabetes, Obesity and Metabolism*, 11, 498–505. <https://doi.org/10.1111/j.1463-1326.2008.00992.x>.

- Catapano, L. A., & Manji, H. K. (2007). G protein-coupled receptors in major psychiatric disorders. *Biochimica et Biophysica Acta*, 1768, 976–993. <https://doi.org/10.1016/j.bbamem.2006.09.025>.
- Cheang, J. Y., & Moyle, P. M. (2018). Glucagon-like peptide-1 (GLP-1)-based therapeutics: Current status and future opportunities beyond type 2 diabetes. *ChemMedChem*, 13, 662–671. <https://doi.org/10.1002/cmdc.201700781>.
- Cromm, P. M., Spiegel, J., & Grossmann, T. N. (2015). Hydrocarbon stapled peptides as modulators of biological function. *ACS Chemical Biology*, 10, 1362–1375. <https://doi.org/10.1021/cb501020r>.
- Deacon, C. F. (2007). Incretin-based treatment of type 2 diabetes: Glucagon-like peptide-1 receptor agonists and dipeptidyl peptidase-4 inhibitors. *Diabetes, Obesity and Metabolism*, 1, 23–31. <https://doi.org/10.1111/j.1463-1326.2007.00765.x>.
- de Graaf, C., Donnelly, D., Wootten, D., Lau, J., Sexton, P. M., Miller, L. J., et al. (2016). Glucagon-like peptide-1 and its class B G protein-coupled receptors: A long march to therapeutic successes. *Pharmacological Reviews*, 68, 954–1013. <https://doi.org/10.1124/pr.115.011395>.
- DeYoung, M. B., MacConell, L., Sarin, V., Trautmann, M., & Herbert, P. (2011). Encapsulation of exenatide in poly-(D,L-lactide-co-glycolide) microspheres produced an investigational long-acting once-weekly formulation for type 2 diabetes. *Diabetes Technology & Therapeutics*, 13, 1145–1154. <https://doi.org/10.1089/dia.2011.0050>.
- Donnelly, D. (2012). The structure and function of the glucagon-like peptide-1 receptor and its ligands. *British Journal of Pharmacology*, 166, 27–41. <https://doi.org/10.1111/j.1476-5381.2011.01687.x>.
- Drucker, D. J., Buse, J. B., Taylor, K., Kendall, D. M., Trautmann, M., Zhuang, D., et al. (2008). Exenatide once weekly versus twice daily for the treatment of type 2 diabetes: A randomised, open-label, non-inferiority study. *The Lancet*, 372, 1240–1250. [https://doi.org/10.1016/S0140-6736\(08\)61206-4](https://doi.org/10.1016/S0140-6736(08)61206-4).
- Drucker, D. J., & Nauck, M. A. (2006). The incretin system: Glucagon-like peptide-1 receptor agonists and dipeptidyl peptidase-4 inhibitors in type 2 diabetes. *The Lancet*, 368, 1696–1705. [https://doi.org/10.1016/S0140-6736\(06\)69705-5](https://doi.org/10.1016/S0140-6736(06)69705-5).
- Dungan, K. M., Povedano, S. T., Forst, T., González, J. G. G., Atisso, C., Sealls, W., et al. (2014). Once-weekly dulaglutide versus once-daily liraglutide in metformin-treated patients with type 2 diabetes (AWARD-6): A randomised, open-label, phase 3, non-inferiority trial. *The Lancet*, 384, 1349–1357. [https://doi.org/10.1016/S0140-6736\(14\)60976-4](https://doi.org/10.1016/S0140-6736(14)60976-4).
- Elbrond, B., Jakobsen, G., Larsen, S., Agerso, H., Jensen, L. B., Rolan, P., et al. (2002). Pharmacokinetics, pharmacodynamics, safety, and tolerability of a single-dose of NN2211, a long-acting glucagon-like peptide 1 derivative, in healthy male subjects. *Diabetes Care*, 25, 1398–1404. <https://doi.org/10.2337/diacare.25.8.1398>.
- Glaesner, W., Mark Vick, A., Millican, R., Ellis, B., Tschang, S., Tian, Y., et al. (2010). Engineering and characterization of the long-acting glucagon-like peptide-1 analogue LY2189265, an Fc fusion protein. *Diabetes/Metabolism Research and Reviews*, 26, 287–296. <https://doi.org/10.1002/dmrr.1080>.
- Hauser, A. S., Chavali, S., Masuho, I., Jahn, L. J., Martemyanov, K. A., Gloriam, D. E., et al. (2018). Pharmacogenomics of GPCR drug targets. *Cell*, 172, 41–54. <https://doi.org/10.1016/j.cell.2017.11.033>.
- Henchey, L. K., Jochim, A. L., & Arora, P. S. (2008). Contemporary strategies for the stabilization of peptides in the α -helical conformation. *Current Opinion in Chemical Biology*, 12, 692–697. <https://doi.org/10.1016/j.cbpa.2008.08.019>.
- Jamieson, A. G., Boutard, N., Sabatino, D., & Lubell, W. D. (2013). Peptide scanning for studying structure-activity relationships in drug discovery. *Chemical Biology & Drug Design*, 81, 148–165. <https://doi.org/10.1111/cbdd.12042>.

- Kim, D., MacConell, L., Zhuang, D., Kothare, P. A., Trautmann, M., Fineman, M., et al. (2007). Effects of once-weekly dosing of a long-acting release formulation of exenatide on glucose control and body weight in subjects with type 2 diabetes. *Diabetes Care*, 30, 1487–1493. <https://doi.org/10.2337/dc06-2375>.
- Knudsen, L. B., Nielsen, P. F., Huusfeldt, P. O., Johansen, N. L., Madsen, K., Pedersen, F. Z., et al. (2000). Potent derivatives of glucagon-like peptide-1 with pharmacokinetic properties suitable for once daily administration. *Journal of Medicinal Chemistry*, 43, 1664–1669. <https://doi.org/10.1021/jm9909645>.
- Kreymann, B., Williams, G., Ghatel, M. A., & Bloom, S. R. (1987). Glucagon-like peptide-1 7-36: A physiological incretin in man. *The Lancet*, 2, 1300–1304. [https://doi.org/10.1016/S0140-6736\(87\)91194-9](https://doi.org/10.1016/S0140-6736(87)91194-9).
- Lam, S., & See, S. (2006). Exenatide: A novel incretin mimetic agent for treating type 2 diabetes mellitus. *Cardiology in Review*, 14, 205–211. <https://doi.org/10.1097/01.crd.0000223655.16253.e4>.
- Larsen, J., Hylleberg, B., Ng, K., & Damsbo, P. (2001). Glucagon-like peptide-1 infusion must be maintained for 24 h/day to obtain acceptable glycemia in type 2 diabetic patients who are poorly controlled on sulphonylurea treatment. *Diabetes Care*, 24, 1416–1421. <https://doi.org/10.2337/diacare.24.8.1416>.
- Lau, J., Bloch, P., Schaffer, L., Pettersson, I., Spetzler, J., Kofoed, J., et al. (2015). Discovery of the once-weekly glucagon-like peptide-1 (GLP-1) analogue semaglutide. *Journal of Medicinal Chemistry*, 58, 7370–7380. <https://doi.org/10.1021/acs.jmedchem.5b00726>.
- Lau, Y. H., de Andrade, P., Wu, Y., & Spring, D. R. (2015). Peptide stapling techniques based on different macrocyclisation chemistries. *Chemical Society Reviews*, 44, 91–102. <https://doi.org/10.1039/C4CS00246F>.
- Lindamood, C. A., & Taylor, J. R. (2015). Emerging new therapies for the treatment of type 2 diabetes mellitus: Glucagon-like peptide-1 receptor agonists. *Clinical Therapeutics*, 37, 483–493. <https://doi.org/10.1016/j.clinthera.2015.01.003>.
- Madsbad, S. (2016). Review of head-to-head comparisons of glucagon-like peptide-1 receptor agonists. *Diabetes, Obesity and Metabolism*, 18, 317–332. <https://doi.org/10.1111/dom.12596>.
- Madsen, K., Knudsen, L. B., Agersoe, H., Nielsen, P. F., Thogersen, H., Wilken, M., et al. (2007). Structure-activity and protraction relationship of long-acting glucagon-like peptide-1 derivatives: Importance of fatty acid length, polarity, and bulkiness. *Journal of Medicinal Chemistry*, 50, 6126–6132. <https://doi.org/10.1021/jm070861j>.
- Manandhar, B., & Ahn, J.-M. (2015). Glucagon-like peptide-1 (GLP-1) analogs: Recent advances, new possibilities, and therapeutic implications. *Journal of Medicinal Chemistry*, 58, 1020–1037. <https://doi.org/10.1021/jm500810s>.
- Marinissen, M. J., & Gutkind, J. S. (2001). G-protein-coupled receptors and signaling networks: Emerging paradigms. *Trends in Pharmacological Sciences*, 22, 368–376. [https://doi.org/10.1016/S0165-6147\(00\)01678-3](https://doi.org/10.1016/S0165-6147(00)01678-3).
- McCarty, D., Coleman, M., & Boland, C. L. (2017). Lixisenatide: A new daily GLP-1 agonist for type 2 diabetes management. *Annals of Pharmacotherapy*, 51, 401–409. <https://doi.org/10.1177/1060028017689878>.
- Meier, J. J. (2012). GLP-1 receptor agonists for individualized treatment of type 2 diabetes mellitus. *Nature Reviews Endocrinology*, 8, 728–742. <https://doi.org/10.1038/nrendo.2012.140>.
- Meier, S., Hucking, K., Ritzel, R., Holst, J. J., Schmiegel, W. H., & Nauck, M. A. (2003). Absence of a memory effect for the insulinotropic action of glucagon-like peptide 1 (GLP-1) in healthy volunteers. *Hormone and Metabolic Research*, 35, 551–556. <https://doi.org/10.1055/s-2003-42658>.
- Meric-Bernstam, F., Saleh, M. N., Infante, J. R., Goel, S., Falchook, G. S., Shapiro, G., et al. (2017). Phase I trial of a novel stapled peptide ALRN-6924 disrupting

- MDMX- and MDM2-mediated inhibition of WT p53 in patients with solid tumors and lymphomas. *Journal of Clinical Oncology*, 35, 2505. https://doi.org/10.1200/JCO.2017.35.15_suppl.2505.
- Muppidi, A., Zou, H., Yang, P.-Y., Chao, E., Sherwood, L., Nunez, V., et al. (2016). Design of potent and proteolytically stable oxyntomodulin analogs. *ACS Chemical Biology*, 11, 324–328. <https://doi.org/10.1021/acscchembio.5b00787>.
- Nauck, M. A., Kleine, N., Ørskov, C., Holst, J. J., Willms, B., & Creutzfeldt, W. (1993). Normalization of fasting hyperglycaemia by exogenous glucagon-like peptide 1 (7–36 amide) in type 2 (non-insulin-dependent) diabetic patients. *Diabetologia*, 36, 741–744. <https://doi.org/10.1007/BF00401145>.
- Oh, D. Y., & Olefsky, J. M. (2016). G protein-coupled receptors as targets for anti-diabetic therapeutics. *Nature Reviews Drug Discovery*, 15, 161–172. <https://doi.org/10.1038/nrd.2015.4>.
- Perry, C. M. (2011). Liraglutide: A review of its use in the management of type 2 diabetes mellitus. *Drugs*, 71, 2347–2373. <https://doi.org/10.2165/11208110-000000000-00000>.
- Persaud, S. J. (2017). Islet G-protein coupled receptors: Therapeutic potential for diabetes. *Current Opinion in Pharmacology*, 37, 24–28. <https://doi.org/10.1016/j.coph.2017.08.001>.
- Pi-Sunyer, X., Astrup, A., Fujioka, K., Greenway, F., Halpern, A., Krempf, M., et al. (2015). A randomized, controlled trial of 3.0 mg of liraglutide in weight management. *The New England Journal of Medicine*, 373, 11–22. <https://doi.org/10.1056/NEJMoa1411892>.
- Pratley, R. E., Nauck, M. A., Barnett, A. H., Feinglos, M. N., Ovalle, F., Harman-Boehm, I., et al. (2014). Once-weekly albiglutide versus once-daily liraglutide in patients with type 2 diabetes inadequately controlled on oral drugs (HARMONY 7): A randomised, open-label, multicentre, non-inferiority phase 3 study. *The Lancet Diabetes & Endocrinology*, 2, 289–297. [https://doi.org/10.1016/S2213-8587\(13\)70214-6](https://doi.org/10.1016/S2213-8587(13)70214-6).
- Rosenstock, J., Raccach, D., Koranyi, L., Maffei, L., Boka, G., Miossec, P., et al. (2013). Efficacy and safety of lixisenatide once daily versus exenatide twice daily in type 2 diabetes inadequately controlled on metformin: A 24-week, randomized, open-label, active-controlled study (GetGoal-X). *Diabetes Care*, 36, 2945–2951. <https://doi.org/10.2337/dc12-2709>.
- Runge, S., Thøgersen, H., Madsen, K., Lau, J., & Rudolph, R. (2008). Crystal structure of the ligand-bound glucagon-like peptide-1 receptor extracellular domain. *The Journal of Biological Chemistry*, 283, 11340–11347. <https://doi.org/10.1074/jbc.M708740200>.
- Scott, L. J. (2014). Liraglutide: A review of its use in adult patients with type 2 diabetes mellitus. *Drugs*, 74, 2161–2174. <https://doi.org/10.1007/s40265-014-0321-6>.
- Szafran, K., Faron-Gorecka, A., Kolasa, M., Kusmider, M., Solich, J., Zurawek, D., et al. (2013). Potential role of G protein-coupled receptor (GPCR) heterodimerization in neuropsychiatric disorders: A focus on depression. *Pharmacological Reports*, 65, 1498–1505. [https://doi.org/10.1016/S1734-1140\(13\)71510-X](https://doi.org/10.1016/S1734-1140(13)71510-X).
- Trujillo, J. M., Nuffer, W., & Ellis, S. L. (2015). GLP-1 receptor agonists: A review of head-to-head clinical studies. *Therapeutic Advances in Endocrinology and Metabolism*, 6, 19–28. <https://doi.org/10.1177/2042018814559725>.
- van Witteloostuijn, S. B., Pedersen, S. L., & Jensen, K. J. (2016). Half-life extension of biopharmaceuticals using chemical methods: Alternatives to PEGylation. *ChemMedChem*, 11, 2474–2495. <https://doi.org/10.1002/cmdc.201600374>.
- Vassilatis, D. K., Hohmann, J. G., Zeng, H., Li, F., Ranchalis, J. E., Mortrud, M. T., et al. (2003). The G protein-coupled receptor repertoires of human and mouse. *Proceedings of the National Academy of Sciences of the United States of America*, 100, 4903–4908. <https://doi.org/10.1073/pnas.0230374100>.
- Verdine, G. L., & Hilinski, G. J. (2012). Stapled peptides for intracellular drug targets. In K. D. Wittrup & G. L. Verdine (Eds.), *Protein engineering for therapeutics, part B* (pp. 3–33). Academic Press. <https://doi.org/10.1016/B978-0-12-396962-0.00001-X>.

- VilSBoll, T., Agerso, H., Krarup, T., & Holst, J. J. (2003). Similar elimination rates of glucagon-like peptide-1 in obese type 2 diabetic patients and healthy subjects. *The Journal of Clinical Endocrinology & Metabolism*, 88, 220–224. <https://doi.org/10.1210/jc.2002-021053>.
- White, A. M., & Craik, D. J. (2016). Discovery and optimization of peptide macrocycles. *Expert Opinion on Drug Discovery*, 11, 1151–1163. <https://doi.org/10.1080/17460441.2016.1245720>.
- Willard, F. S., Bueno, A. B., & Sloop, K. W. (2012). Small molecule drug discovery at the glucagon-like peptide-1 receptor. *Experimental Diabetes Research*, 2012, 1–9. <https://doi.org/10.1155/2012/709893>.
- Wysham, C., Blevins, T., Arakaki, R., Colon, G., Garcia, P., Atisso, C., et al. (2015). Efficacy and safety of dulaglutide added onto pioglitazone and metformin versus exenatide in type 2 diabetes in a randomized controlled trial (AWARD-1). *Diabetes Care*, 37, 2159–2167. <https://doi.org/10.2337/dc13-2760>.
- Yang, P.-Y., Zou, H., Chao, E., Sherwood, L., Nunez, V., Keeney, M., et al. (2016). Engineering a long-acting, potent GLP-1 analog for microstructure-based transdermal delivery. *Proceedings of the National Academy of Sciences of the United States of America*, 113, 4140–4145. <https://doi.org/10.1073/pnas.1601653113>.
- Yang, P.-Y., Zou, H., Lee, C., Muppidi, A., Chao, E., Fu, Q., et al. (2018). Stapled, long-acting glucagon-like peptide 2 analog with efficacy in dextran sodium sulfate induced mouse colitis models. *Journal of Medicinal Chemistry*, 61, 3218–3223. <https://doi.org/10.1021/acs.jmedchem.7b00768>.